Tutorial Problem Set 1

Exercise 1. Show that each of the following functions $f(z)$ satisfies the Cauchy Riemann equations at the origin, but $f'(0)$ does not exist:

(a) $\begin{cases} \frac{xy^2(x+iy)}{x^2+y^4}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

(b) $\begin{cases} \frac{x^3y(y-ix)}{x^6+y^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

Exercise 2. Show that $f(z) = xy + iy^2$ is differentiable at the origin, but nowhere analytic.

Exercise 3. Show that $f(z) = \cos x + i \sinh y$ is differentiable only at $P(\pi/2, 0)$, and find $f'(\pi/2, 0)$. Is f analytic at P?

Exercise 4. Show that $f(z) = (2x^2 + y) + i(y^2 - x)$ is differentiable along the straight line $y = 2x$, and find the corresponding derivative. What about its analyticity?

Exercise 5. Show that $f(z) = x^2 + y^2 + 2xyi$ is nowhere analytic but is differentiable along the the real axis.

Exercise 6. Show that each of the following functions $f(z)$ is entire, and find the derivative at each point z of the complex plane:

(a) $4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$,

(b) $f(z) = \cos x \cosh y - i \sin x \sinh y$,

(c) $f(z) = 3(x^2 - y^2) + 5x + i[(6x + 5)y - 6]$.

Exercise 7. Find the choice of the constants (*mentioned in brackets*) such that the following functions are entire:

(a) $f(z) = (ax + y) + \mathrm{i}(3y - bx), \quad (a, b)$

(b) $f(z) = x^2 + axy + by^2 + \mathrm{i}(cx^2 + dxy + y^2), \quad (a, b, c, d)$

Exercise 8. Show that $f(z) = \frac{x-1}{(x-1)^2+y^2} - i\frac{y}{(x-1)^2+y^2}$ is analytic in an appropriate domain, and find its derivative in the domain.